
CS 421 --- Ambiguous Grammars / First and Follow Sets

Name	Netid

Please write your name/netid legibly in dark ink. Hand in one copy per team. Do not staple or mangle the corners.

Part 1 --- Ambiguous Grammars

Consider the following ambiguous grammar.

Example 1)

$$\begin{array}{l} E \rightarrow E + E \\ \quad | E * E \\ \quad | - E \\ \quad | i \end{array}$$

Problem 1) Prove that this grammar is ambiguous by giving an input with two parse trees.

Problem 2) The instructor will give you a version of this grammar that is not ambiguous. We will let $*$ and $+$ have their usual precedences, and associate to the left. Unary minus binds most tightly. Show that your input has only one parse tree now.

$$\begin{array}{l} E \rightarrow E + F \mid F \\ F \rightarrow F * U \mid U \\ U \rightarrow - U \mid i \end{array}$$

Problem 3) Stratification removed the ambiguities. How did it do that?

Here is a new grammar for you.

$$\begin{array}{l} E \rightarrow E a E \\ \quad | E b E \\ \quad | E x E \\ \quad | E y E \\ \quad | i \end{array}$$

Problem 4) Prove that this grammar is ambiguous by giving an input that has two parse trees.

Problem 5) Stratify the grammar to remove ambiguity. Let a have highest precedence, associating to the left. Next is b , associating to the left. Then we have y associating to the right. Let x have the lowest precedence, and associate to the right. Show that your sentence has only one parse tree.

Part 2 --- First and Follow Sets

Calculate the first and follow sets for these grammars.

Example 2)

$$\begin{array}{l} S \rightarrow a E b \\ \quad | \quad x \\ E \rightarrow x y \\ \quad | \quad \epsilon \end{array}$$

Problem 6)

$$\begin{array}{l} S \rightarrow a E F b \\ \quad | \quad x E F \\ E \rightarrow x y \\ \quad | \quad \epsilon \\ F \rightarrow E z q \\ \quad | \quad w S \end{array}$$

Problem 7)

$$\begin{array}{l} S \rightarrow a F E b \\ \quad | \quad x F E \\ E \rightarrow x y \\ \quad | \quad \epsilon \\ F \rightarrow E z q \\ \quad | \quad w S \end{array}$$